

11.15 Taking partial derivative of the Hamiltonian function w.r.t r_i ,

$$\frac{\partial H}{\partial r_i} = \frac{\partial E(\mathbf{z})}{\partial r_i} + \frac{\partial K(\mathbf{r})}{\partial r_i}$$

As per (11.54), the potential energy is a function of $\mathbf{z}$, and is independent of momentum variable $\mathbf{r}$, giving us:

$$\begin{aligned}\frac{\partial H}{\partial r_i} &= 0 + \frac{\partial (1/2 \sum_i r_i^2)}{\partial r_i} \\ &= r_i\end{aligned}$$

As per (11.53),

$$\begin{aligned}r_i &= \frac{dz_i}{d\tau} \\ \Rightarrow \frac{dz_i}{d\tau} &= \frac{\partial H}{\partial r_i}\end{aligned}$$

This proves (11.58) and (11.53) are equivalent.

Similarly, taking partial derivative of the Hamiltonian function w.r.t z_i ,

$$\frac{\partial H}{\partial z_i} = \frac{\partial E(\mathbf{z})}{\partial z_i} + \frac{\partial K(\mathbf{r})}{\partial z_i}$$

As per (11.56), the kinetic energy is a function of $\mathbf{r}$, and is independent of position variable $\mathbf{z}$, giving us:

$$\frac{\partial H}{\partial z_i} = \frac{\partial E(\mathbf{z})}{\partial z_i} + 0$$

As per (11.55),

$$\begin{aligned}\frac{\partial E(\mathbf{z})}{\partial z_i} &= -\frac{dr_i}{d\tau} \\ \Rightarrow \frac{dr_i}{d\tau} &= -\frac{\partial H}{\partial z_i}\end{aligned}$$

This proves (11.59) and (11.55) are equivalent.